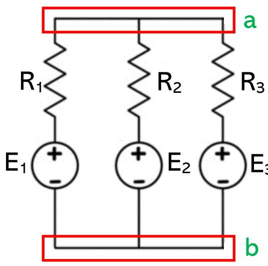
DC Circuit Part 4

Millman's Theorem

Continuation....

Millman's circuit equivalent:



The common node voltage between the parallel branches is given by:

$$V_{ab} = \frac{\frac{E_1}{R_1} + \frac{E_2}{R_2} + \frac{E_3}{R_3}}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}}$$

Reciprocity Theorem

"If an emf in a circuit A produces a current in circuit B, then the same emf in circuit B produces the same current in circuit A."

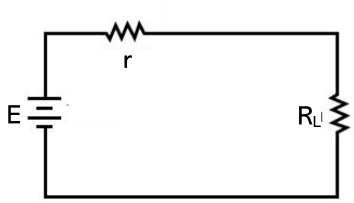
Compensation Theorem

"Any resistance R in a branch of a network in which a current I is flowing can be replaced for the purpose of calculation, by a voltage equal to IR."

Maximum Power Transfer Theorem

"The maximum power transferred to a load resistor occurs when it has a value equal to the resistance of the network looking back from the load terminals with all the sources of voltage replaced by their internal resistances."

Under the condition of maximum power transfer, the efficiency is only 50%.



When $R_L = r$, power delivered to the load

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TRANSFORMATIONS or CONVERSIONS

Delta (Δ) to Wye (Y) or Pi to T

"If the delta is to be electrically equivalent to the star or wye, the resistance between any pair of terminals on the delta must be equal to the resistance of the corresponding terminals on the star or wye."

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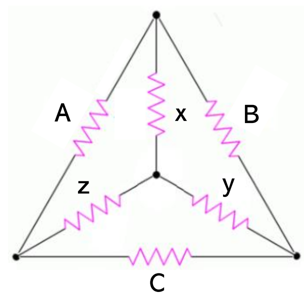

$$x = \frac{AB}{A+B+C} \qquad y = \frac{BC}{A+B+C}$$

$$z = \frac{AC}{A+B+C}$$

In general,

$$R_Y = \frac{\text{Product of adjacent R in } \Delta}{\sum \text{ of all R in } \Delta}$$

Wye (Y) to Delta (Δ) T to Pi

$$A = \frac{xy+yz+xz}{y} \qquad B = \frac{xy+yz+xz}{z}$$

$$C = \frac{xy+yz+xz}{x}$$

In general,

$$R_\Delta = \frac{\sum \text{ of cross products in Y}}{\text{Opposite R in Y}}$$

Glossary 1

Ammeter An instrument for measuring the flow of electrical current in amperes. Ammeters are always connected in series with the circuit to be tested.

Ampacity The maximum amount of electric current a conductor or device can carry before sustaining immediate or progressive deterioration.

Ampere-Hour (Ah) A unit of measure for battery capacity. It is obtained by multiplying the current (in amperes) by the time (in hours) during which current flows. For example, a battery which provides 5 amperes for 20 hours is said to deliver 100 ampere - hours.

Ampere (A) A unit of measure for the intensity of an electric current flowing in a circuit. One ampere is equal to a current flow of one coulomb per second.

Apparent Power Measured in volt-amperes (VA). Apparent power is the product of the rms voltage and the rms current.

Capacitance The ability of a body to store an electrical charge. Measured in farads as the ratio of the electric charge of the object (Q, measured in coulombs) to the voltage across the object (V, measured in volts).

Capacitor A device used to store an electric charge, consisting of one or more pairs of conductors separated by an insulator. Commonly used for filtering out voltage spikes.

Circuit A closed path in which electrons from a voltage or current source flow. Circuits can be in series, parallel, or in any combination of the two.

Circuit Breaker An automatic device for stopping the flow of current in an electric circuit. To restore service, the circuit breaker must be reset (closed) after correcting the cause of the overload or failure. Circuit breakers are used in conjunction with protective relays to protect circuits from faults.

Conductor Any material where electric current can flow freely. Conductive materials, such as metals, have a relatively low resistance. Copper and aluminum wire are the most common conductors.

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